

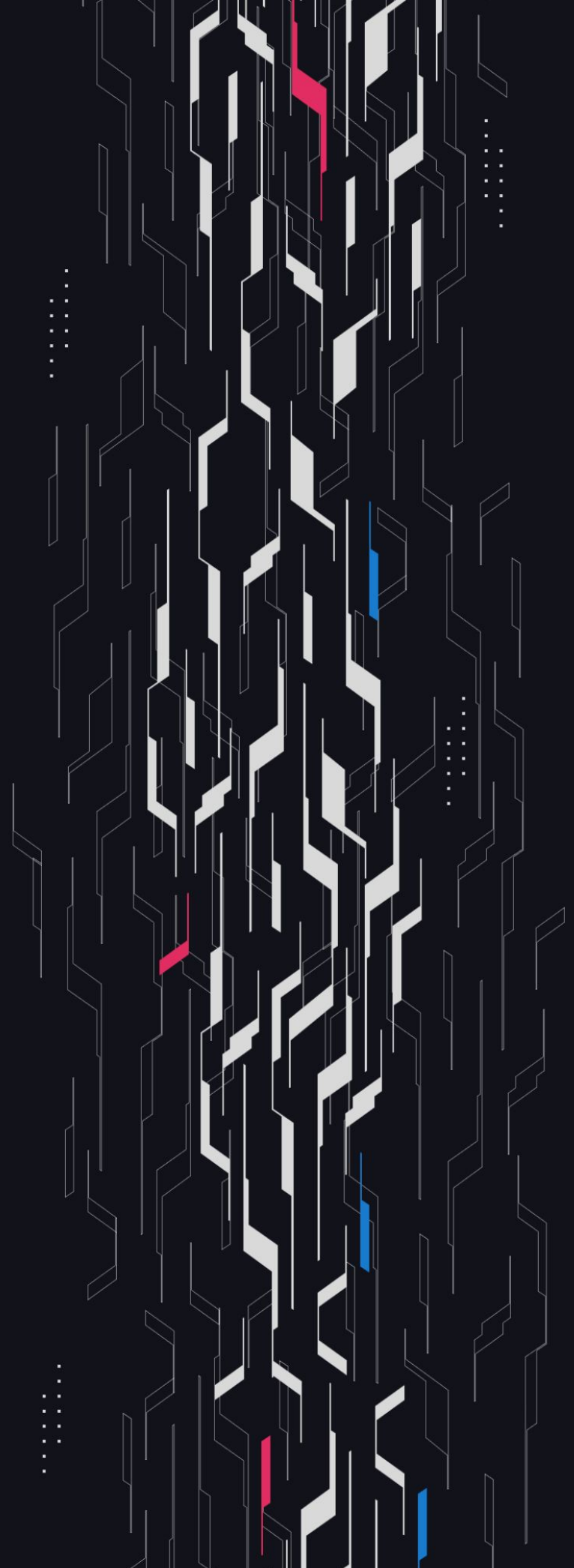
GA GUARDIAN

M0

**Liquidity Delivery
Updates**

Security Assessment

June 1st, 2026



Summary

Audit Firm Guardian

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Client Firm M0

Final Report Date June 1, 2026

Audit Summary

M0 engaged Guardian to review the Liquidity Delivery updates in scope. From May 19 through May 26, 2026, two Guardian auditors reviewed the pull request changes affecting EVM and SVM order-book behavior, cancellation flows, and liquidity delivery execution paths.

Confidence Ranking

Given the lack of High and Critical issues detected during the main review, Guardian assigns a Confidence Ranking of 5 to the protocol. Guardian advises the protocol to consider periodic review with future changes. For detailed understanding of the Guardian Confidence Ranking, please see the rubric on the following page.

✓ Verify the authenticity of this report on Guardian's GitHub: <https://github.com/guardianaudits>

Guardian Confidence Ranking

Confidence Ranking	Definition and Recommendation	Risk Profile
5: Very High Confidence	Codebase is mature, clean, and secure. No High or Critical vulnerabilities were found. Follows modern best practices with high test coverage and thoughtful design. Recommendation: Code is highly secure at time of audit. Low risk of latent critical issues.	0 High/Critical findings and few Low/Medium severity findings.
4: High Confidence	Code is clean, well-structured, and adheres to best practices. Only 1 Significant issue was uncovered per week. Design patterns are sound, and test coverage is strong. Recommendation: Suitable for deployment after remediations; consider periodic review with changes.	0-1 High/Critical findings per engagement week and little to no Medium severity issues. Varied Low severity findings.
3: Moderate Confidence	Medium-severity and occasional High-severity issues found. Code is functional, but there are concerning areas (e.g., weak modularity, risky patterns). No critical design flaws, though some patterns could lead to issues in edge cases. Recommendation: Address issues thoroughly and consider a targeted follow-up audit depending on code changes.	1-2 High/Critical findings per engagement week.
2: Low Confidence	Code shows frequent emergence of Critical/High vulnerabilities. Audit revealed recurring anti-patterns, weak test coverage, or unclear logic. These characteristics suggest a high likelihood of latent issues. Recommendation: Post-audit development and a second audit cycle are strongly advised.	2-4 High/Critical findings per engagement week. Or additional High/Critical findings uncovered in remediation review which have not been resolved and confirmed by Guardian.
1: Very Low Confidence	Code has systemic issues. Multiple High/Critical findings (≥ 5/week), poor security posture, and design flaws that introduce compounding risks. Safety cannot be assured. Recommendation: Halt deployment and seek a comprehensive re-audit after substantial refactoring.	≥ 5 High/Critical findings and overall systemic flaws.

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Project Overview

Project Summary

Project Name	M0
Codebase	https://github.com/m0-foundation/liquidity-delivery
Commit(s)	Main Review: 4ad31c9b13a36ef83a58a7af8a7d01510864f51a Remediation: 22475fc629b598bf5e67a1e5ee9a66b33f657b10

Audit Summary

Delivery Date	June 1, 2026
Audit Methodology	Static Analysis, Manual Review

Vulnerability Summary

Vulnerability Level	Total	Pending	Declined	Acknowledged	Partially Resolved	Resolved
● Critical	0	0	0	0	0	0
● High	0	0	0	0	0	0
● Medium	0	0	0	0	0	0
● Low	1	0	0	0	0	1
● Info	3	0	0	1	0	2

Audit Scope & Methodology

The following PR is in scope: <https://github.com/m0-foundation/liquidity-delivery/pull/72>

Audit Scope & Methodology

Vulnerability Classifications

Severity	Impact: <i>High</i>	Impact: <i>Medium</i>	Impact: <i>Low</i>
Likelihood: <i>High</i>	● Critical	● High	● Medium
Likelihood: <i>Medium</i>	● High	● Medium	● Low
Likelihood: <i>Low</i>	● Medium	● Low	● Low

Impact

- High** Significant loss of assets in the protocol, significant harm to a group of users, or a core functionality of the protocol is disrupted.
- Medium** A small amount of funds can be lost or ancillary functionality of the protocol is affected. The user or protocol may experience reduced or delayed receipt of intended funds.
- Low** Can lead to any unexpected behavior with some of the protocol's functionalities that is notable but does not meet the criteria for a higher severity.

Likelihood

- High** The attack is possible with reasonable assumptions that mimic on-chain conditions, and the cost of the attack is relatively low compared to the amount gained or the disruption to the protocol.
- Medium** An attack vector that is only possible in uncommon cases or requires a large amount of capital to exercise relative to the amount gained or the disruption to the protocol.
- Low** Unlikely to ever occur in production.

Audit Scope & Methodology

Methodology

Guardian is the ultimate standard for Smart Contract security. An engagement with Guardian entails the following:

- Two competing teams of Guardian security researchers performing an independent review.
- A dedicated fuzzing engineer to construct a comprehensive stateful fuzzing suite for the project.
- An engagement lead security researcher coordinating the 2 teams, performing their own analysis, relaying findings to the client, and orchestrating the testing/verification efforts.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross-referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.
Comprehensive written tests as a part of a code coverage testing suite.
- Contract fuzzing for increased attack resilience.

Round One Findings & Resolutions

ID	Title	Category	Severity	Status
I-01	NatSpec Comments Are Stale	Documentation	● Info	Resolved
I-02	Gasless Cancel Cannot Be Wrapped	Informational	● Info	Resolved
I-03	Filling Native Orders May Fail	Warning	● Info	Acknowledged

I-01 | NatSpec Comments Are Stale

Category	Severity	Location	Status
Documentation	● Info	IOrderBook.sol: 349-351,363-365,382-384	Resolved

Description

Some IOrderBook NatSpec comments are stale after recent order-opening changes. Order . sender and OrderData . sender are documented as the address that provided funds, but sender is now the order owner while the funder/msg . sender provides tokens. The cancelOrder () fee comments also refer to same-chain and cross-chain fills even though the section documents cancellations. Finally, OrderCompleted is documented as destination-chain-only, but it can also be emitted by reportFill () on the origin chain.

Recommendation

Update the NatSpec comments to describe sender as the order owner, distinguish it from funder, replace fill wording with cancel wording in cancelOrder () documentation, and clarify where OrderCompleted may be emitted.

Resolution

M0 Team: Resolved.

I-02 | Gasless Cancel Cannot Be Wrapped

Category	Severity	Location	Status
Informational	● Info	OrderBook.sol: 242-246	Resolved

Description

The removed `cancelOrderFor()` functionality cannot be replicated by a generic wrapper in the same way as gasless `openOrder()` flows. Before the deadline, `cancelOrder()` authorizes cancellation by checking that `msg.sender` is the order recipient or, for same-chain orders, the order sender. If a wrapper verifies a user signature and calls `cancelOrder()`, `OrderBook` sees the wrapper as `msg.sender`, so the call fails unless the wrapper itself is the authorized address.

Setting the wrapper as `recipient` is not equivalent because fills transfer `tokenOut` directly to `recipient` without executing a callback. The wrapper would become a custody/accounting layer and would need additional forwarding, claiming, or off-chain keeper logic to route filled funds to the actual user.

Recommendation

Confirm this behavior is intentional. If not, consider restoring gasless cancellation support in `OrderBook`.

Resolution

M0 Team: Resolved.

I-03 | Filling Native Orders May Fail

Category	Severity	Location	Status
Warning	● Info	fill.rs: 82-159	Acknowledged

Description

After compiling the orderbook, the `FillNativeOrder` tests should be checked for success.

Currently, the plan is to compile the program using Solana v2.1.0 and Anchor v0.31.1. However, for different versions of the toolchain, the following issue may materialize:

The SVM `FillNativeOrder` account context is too large for SBF stack limits. During `anchor build`, the compiler reports that the generated `FillNativeOrder::try_accounts()` frame uses a stack offset of 4112, exceeding the 4096 byte maximum. The generated parser then traps at runtime before `handler()` can execute successfully. Any native same-chain `fill_native_order()` call fails with `ProgramFailedToComplete` and an access violation.

As a result, SVM same-chain orders cannot be filled. Solvers cannot deliver the recipient's `token_out` and receive the escrowed `token_in`; users are forced into cancellation/refund paths, and same-chain liquidity delivery is unavailable on SVM.

Recommendation

Make sure the right toolchain was used and the tests are passing before deploying the program.

Resolution

M0 Team: Acknowledged.

Remediation Findings & Resolutions

ID	Title	Category	Severity	Status
L-01	Cancel TYPEHASH Diverges From EIP-712	Standards Compliance	Low	Resolved

L-01 | Cancel TYPEHASH Diverges From EIP-712

Category	Severity	Location	Status
Standards Compliance	● Low	OrderBook.sol: 59-60	Resolved

Description

The preimage of the CANCEL_ORDER_TYPEHASH includes whitespaces after the commas that separate each struct element.

```
/// @dev keccak256("CancelOrder(bytes32 orderId, address bridgeAdapter, bytes
bridgeAdapterArgs)")
bytes32 public constant CANCEL_ORDER_TYPEHASH = 0x69 ...
1b5b4e13b800c6d41c4792cfc2a12d0bd9ad19da6e0bfe8ac04f;
```

This implementation is not true to the EIP-712 specification, where the whitespaces are omitted.

Recommendation

Consider using EIP-712 compliant digest if it won't break already existing integrators.

Resolution

M0 Team: Resolved.

Disclaimer

This report is not, nor should be considered, an “endorsement” or “disapproval” of any particular project or team. This report is not, nor should be considered, an indication of the economics or value of any “product” or “asset” created by any team or project that contracts Guardian to perform a security assessment. This report does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors, business, business model or legal compliance.

This report should not be used in any way to make decisions around investment or involvement with any particular project. This report in no way provides investment advice, nor should be leveraged as investment advice of any sort. This report represents an extensive assessing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Guardian’s position is that each company and individual are responsible for their own due diligence and continuous security. Guardian’s goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies, and in no way claims any guarantee of security or functionality of the technology we agree to analyze.

The assessment services provided by Guardian is subject to dependencies and under continuing development. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis. Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives, and other unpredictable results. The services may access, and depend upon, multiple layers of third-parties.

Notice that smart contracts deployed on the blockchain are not resistant from internal/external exploit. Notice that active smart contract owner privileges constitute an elevated impact to any smart contract’s safety and security. Therefore, Guardian does not guarantee the explicit security of the audited smart contract, regardless of the verdict.

About Guardian

Founded in 2022 by DeFi experts, Guardian is a leading audit firm in the DeFi smart contract space. With every audit report, Guardian upholds best-in-class security while achieving our mission to relentlessly secure DeFi.

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